Zhike Peng.

Ph.D. Student in Computer Science | Tsinghua University

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Research interests. My interests span theoretical computer science and artificial intelligence, with a particular focus on mechanism design and blockchain. I am also interested in graph theory, reinforcement learning theory, and world models.

Education

Tsinghua University | Department of Computer Science and Technology Sep 2026 - Present

Ph.D. in Computer Science and Technology

Advisor: Professor Jing Chen.

Tsinghua University | Zhili College Sep 2022 - Jun 2026

Bachelor's degree in Information and Computing Science

GPA: 3.86. Comprehensive Excellence Scholarship (2024, 2025); Social Work Scholarship (2023).

Publications

Uniform-Price Mechanism Design for Automated Market Makers

Zhike Peng.

IJTCS-FAW 2026 | Accepted

arXiv | PDF

ScenePhotographer: Object-Oriented Photography for Residential Scenes

Shao-Kui Zhang, Hanxi Zhu, Xuebin Chen, Jinghuan Chen, **Zhike Peng**, Ziyang Chen, Yong-Liang Yang, Song-Hai Zhang.

ACM Multimedia 2024 | Oral presentation

DOI | PDF | Code

Research experience

Uniform-price mechanism design for automated market makers 2025 - 2026

Tsinghua University | Jing Chen's group | Undergraduate thesis

Designed uniform-price mechanisms for automated market makers and analyzed their economic properties and social efficiency.

Disentangling visual scene representations 2025

Tsinghua University | Jun Zhu's group | Research course project

Combined 3D VAEs and Slot Attention to investigate representation learning, feature disentanglement, and semantic segmentation in 3D scenes.

Image reconstruction from EEG signals 2024 - 2025

Tsinghua Information Retrieval Lab | Research intern

Explored reconstructing visual images from EEG signals using VQGAN and GPT.

Automatic viewpoint selection in 3D scenes 2023 - 2024

Tsinghua University | Song-Hai Zhang's group | Student Research Training (SRT) project

Studied automatic camera selection in virtual 3D scenes using object relationships, scene information, and composition criteria. Received the SRT Outstanding Project Second Prize.

Industry experience

Machine Learning 2025 - 2026

Huawei | HiSilicon Turing

Worked on fundamental AI algorithms and AI product development, including algorithm design, experimental validation, and product exploration.

Large Language Model Research 2025

Zhipu AI

Investigated reinforcement learning to improve multi-turn tool use in large language models, with experiments on training and evaluation.

Honors & awards

2025 **Third Place** | Tsinghua-NTHU-HKUST Programming Contest

2024 **Gold Medal | 9th Place** | ICPC Shanghai Regional

2024 **Outstanding Project | Second Prize** | Tsinghua Student Research Training (SRT) Program

2023 **Champion** | CCPC Qinhuangdao Regional

2023 **Gold Medal | 14th Place** | ICPC Xi'an Regional

2023 **First Prize** | Tsinghua-NTHU-HKUST Programming Contest

2023 **Beijing First Prize** | China Undergraduate Mathematical Contest in Modeling

2022 **Gold Medal | 4th Place** | CCPC Guangzhou Regional

2022 **Gold Medal | 16th Place** | ICPC Hangzhou Regional

2022 **Silver Medal | 12th Place** | ICPC Hong Kong Regional

2021 **Silver Medal | 51st Place** | CCF National Olympiad in Informatics (NOI)

Tsinghua University: Comprehensive Excellence Scholarship (2024, 2025); Social Work Scholarship (2023).

Leadership & service

Student Counselor During Ph.D. studies

Wuqiong College, Tsinghua University

Support student education and day-to-day student affairs.

Vice President, Zhili College TMS During undergraduate studies

Tsinghua University

Selected for the 12th cohort of the Lin Feng Program. Served as a TMS staff member, head of its theory department, and vice president; also served as a class arts representative and party-study group leader.

Community engagement

Aug 2023 | Jining, Shandong - Summer volunteer teaching and academic exchange with local secondary-school students.

Jan 2024 | Shibadong Village, Hunan - Field research on targeted poverty alleviation and rural revitalization.

Aug 2024 | Shanxi - Visits to energy and manufacturing enterprises to study industrial transformation and upgrading.

Aug 2024 | Beipiao, Liaoning - Field research on the revitalization and industrial upgrading of an established industrial city.